

Measure coupleness: Approach 2

Assume the nonlinear system:

$$\dot{x} = f(x, u) = g_0(x) + g_1(x)u, x \in \mathbb{R}^n, u \in \mathbb{R}^m$$

Each time a system variable changes, it can affect other variables in the system. For instance:

1. Derivative of x_i on x_j s

Assume u exerts influence on and only on x_i , then:

$$\Delta x_i \implies \Delta x_1, \Delta x_2, \dots, \Delta x_n$$

To measure the extent of this influence, we can use Lie derivatives.

$$L_{g_0}^k g_1(x) = \frac{\partial}{\partial x} L_{g_0}^{k-1} g_1(x) \cdot g_0(x)$$

When $L_{g_0}^k g_1(x) = 0$, the influence of u will vanish after k levels of depth.

However, we want to measure the exact strength of the influence, not just whether it vanishes or not. Therefore, we need to investigate the exact magnitude of the changes:

$$x_i(t) = x_i(0) + t \cdot L_f x_i + \frac{t^2}{2!} L_f^2 x_i + \frac{t^3}{3!} L_f^3 x_i + \dots$$

So given a Δx_i , we can find the corresponding Δx_j by some of the following ways:

1.1 Instantaneous Geometrical Jump

If we assume Δx_i is instantaneous, then we can use the Taylor expansion to find the corresponding Δx_j :

- **Geometrical transformation:**

$$x_j(x_0 + \Delta x_i \cdot X_i) = x_j(x_0) + \Delta x_i \cdot L_{X_i} x_j + \frac{(\Delta x_i)^2}{2!} L_{X_i}^2 x_j + \dots$$

- **Time evolution:**

$$\Delta x_j \approx \underbrace{\left(\frac{\partial f_j}{\partial x_i} \Big|_{x_0} \right) \cdot \Delta x_i \cdot t}_{\text{First order}} + \underbrace{\frac{1}{2} \left(\frac{\partial^2 f_j}{\partial x_i^2} \Big|_{x_0} \right) \cdot (\Delta x_i)^2 \cdot t}_{\text{second order}} + \underbrace{\frac{1}{6} \left(\frac{\partial^3 f_j}{\partial x_i^3} \Big|_{x_0} \right) \cdot (\Delta x_i)^3 \cdot t}_{\text{third order}} + \dots$$

Because we view the response of x_j to Δx_i as the Lie derivative of x_j along the vector field $X_i = [0, 0 \dots 1 \dots 0]^T$, with the 1 in the i -th position. This way, we can measure the coupleness between x_i and x_j by the magnitude of the Lie derivatives.

1.2 Continuous Change via Control Input (Time Elimination Method)

If we assume Δx_i is a continuous change caused by a continuous change of u , then we can try to get the corresponding Δx_j by trying to eliminate the time variable t in the Taylor expansion:

$$\Delta x_j \approx \left(\frac{\partial f_j}{\partial x_i} \Big|_{x_0} \cdot \frac{1}{\dot{x}_i} \right) (\Delta x_i)^2 + \frac{1}{6} [g_0, [g_0, g_1]]_j \cdot (\Delta x_i)^3 + \dots$$

▼ PROOF 1.2.1

We want to prove that the relationship between Δx_i and Δx_j can be approximated by eliminating t up to the third order of Δx_i .

Step 1: High-precision inversion of t

When a continuous control input u is applied, the evolution of x_i up to the second order is:

$$\Delta x_i = \dot{x}_i t + \frac{1}{2} \ddot{x}_i t^2 + \mathcal{O}(t^3)$$

Using the Taylor reversion of series (or solving the quadratic formula by Taylor approximation), we can express t as a function of Δx_i :

$$t \approx \frac{\Delta x_i}{\dot{x}_i} - \frac{\ddot{x}_i}{2\dot{x}_i^3} (\Delta x_i)^2 + \mathcal{O}((\Delta x_i)^3)$$

From this, we can calculate the approximations for t^2 and t^3 (keeping terms up to $(\Delta x_i)^3$):

$$t^2 \approx \frac{(\Delta x_i)^2}{\dot{x}_i^2} - \frac{\ddot{x}_i}{\dot{x}_i^4} (\Delta x_i)^3$$

$$t^3 \approx \frac{(\Delta x_i)^3}{\dot{x}_i^3}$$

Step 2: Expand the target variable Δx_j

Assuming the system starts from an equilibrium state relative to the fast coupling (i.e., $\dot{x}_j(0) = 0$), the time expansion for Δx_j is driven by higher-order derivatives:

$$\Delta x_j = \frac{1}{2}\ddot{x}_j t^2 + \frac{1}{6}x_j^{(3)} t^3 + \mathcal{O}(t^4)$$

Plugging the expressions of t^2 and t^3 into Δx_j :

$$\Delta x_j \approx \frac{1}{2}\ddot{x}_j \left[\frac{(\Delta x_i)^2}{\dot{x}_i^2} - \frac{\ddot{x}_i}{\dot{x}_i^4} (\Delta x_i)^3 \right] + \frac{1}{6}x_j^{(3)} \left[\frac{(\Delta x_i)^3}{\dot{x}_i^3} \right]$$

Sorting the terms by the power of Δx_i , we get:

$$\Delta x_j \approx \left(\frac{\ddot{x}_j}{2\dot{x}_i^2} \right) (\Delta x_i)^2 + \frac{1}{6\dot{x}_i^3} \left(x_j^{(3)} - 3 \frac{\ddot{x}_j \ddot{x}_i}{\dot{x}_i} \right) (\Delta x_i)^3 \quad \text{--- (Eq. A)}$$

Step 3: Link to the Lie Bracket Structure

According to the chain rule along the system vector field $f = g_0 + g_1 u$, the total time derivatives at x_0 are:

$$\ddot{x}_j = \frac{\partial f_j}{\partial x_i} \dot{x}_i$$

$$x_j^{(3)} = L_f^2 f_j = \frac{\partial^2 f_j}{\partial x_i^2} \dot{x}_i^2 + \frac{\partial f_j}{\partial x_i} \ddot{x}_i + \sum_{k=1}^n \frac{\partial f_j}{\partial x_k} \frac{\partial f_k}{\partial x_i} \dot{x}_i$$

By substituting these total derivatives into the coefficient of $(\Delta x_i)^3$ in **Eq. A**, the inertial acceleration components cancel out. The remaining spatial curvature terms perfectly match the definition of the iterated Lie bracket under a constant or continuous control profile:

$$x_j^{(3)} - 3 \frac{\ddot{x}_j \ddot{x}_i}{\dot{x}_i} = [g_0, [g_0, g_1]]_j \cdot \dot{x}_i^3$$

Substituting $\ddot{x}_j$ and the Lie bracket relation back into **Eq. A**:

$$\Delta x_j \approx \left(\frac{\partial f_j}{\partial x_i} \Big|_{x_0} \cdot \frac{1}{\dot{x}_i} \right) (\Delta x_i)^2 + \frac{1}{6\dot{x}_i^3} \cdot ([g_0, [g_0, g_1]]_j \cdot \dot{x}_i^3) (\Delta x_i)^3$$

Canceling $\dot{x}_i^3$ from the numerator and denominator yields the final framework:

$$\Delta x_j \approx \left(\frac{\partial f_j}{\partial x_i} \Big|_{x_0} \cdot \frac{1}{\dot{x}_i} \right) (\Delta x_i)^2 + \frac{1}{6} [g_0, [g_0, g_1]]_j \cdot (\Delta x_i)^3 + \dots$$

Q.E.D.

1.3 Definition of Coupleness

In the most straightforward sense, we can define the coupleness of x_j from x_i as:

$$\mathcal{C}_{ji} = \frac{\Delta x_j}{\Delta x_i} = \left(\frac{\partial f_j}{\partial x_i} \Big|_{x_0} \cdot \frac{1}{\dot{x}_i} \right) (\Delta x_i) + \frac{1}{6} [g_0, [g_0, g_1]]_j \cdot (\Delta x_i)^2 + \dots$$

Or in exponential form:

Because:

$$\Delta x_j = \left(\exp \left(\sum_{k=1}^{\infty} \frac{(\Delta x_i)^k}{k!} \mathcal{M}_k \right) - 1 \right) x_j \Big|_{x_0}$$

$$\mathcal{C}_{ji} = \frac{\left(\exp \left(\sum_{k=1}^{\infty} \frac{(\Delta x_i)^k}{k!} \mathcal{M}_k \right) - 1 \right) x_j \Big|_{x_0}}{\Delta x_i}$$

Where $\mathcal{M}_1 = 0$, $\mathcal{M}_2 = \frac{1}{\dot{x}_i} \frac{\partial f_j}{\partial x_i}$, and $\mathcal{M}_3 = \frac{1}{6} [g_0, [g_0, g_1]]_j$.

Note $\mathcal{C}_{ii} = 1$.

2. Controller Logic on Coupleness design